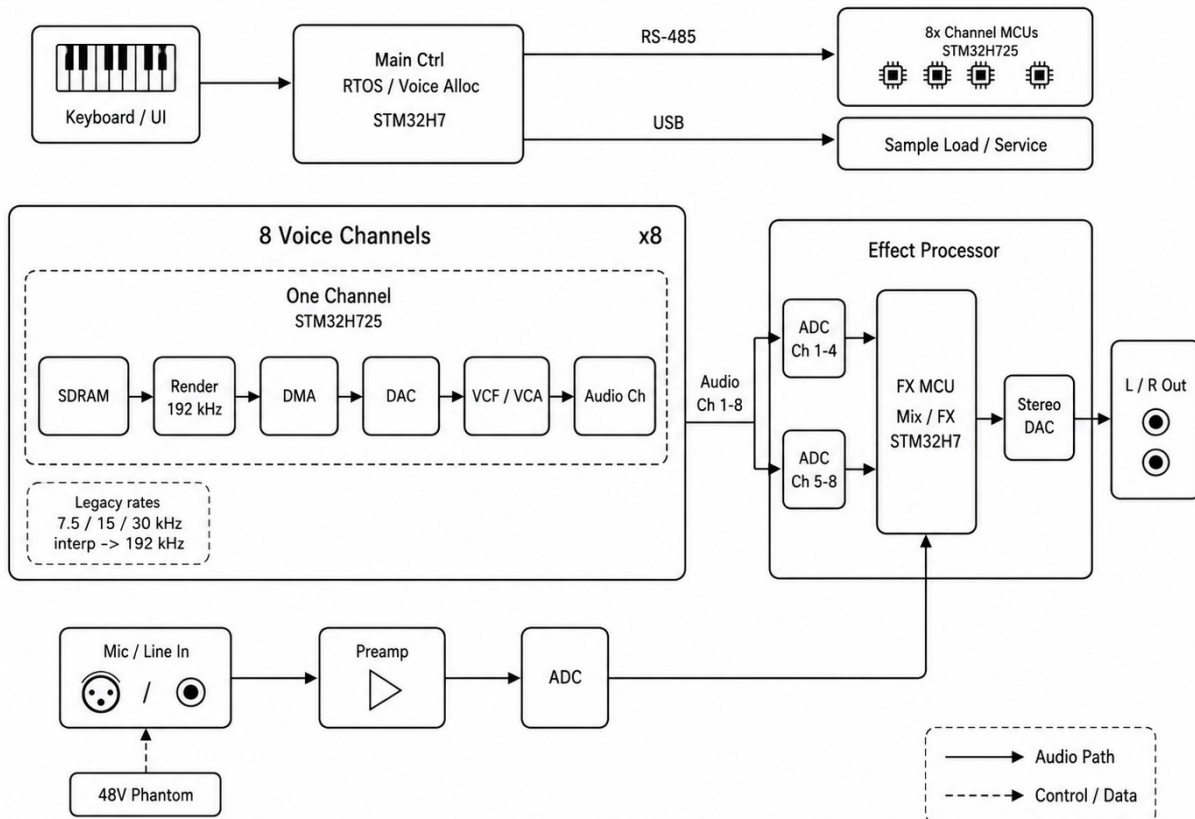


Synthesizer Channel & Effects System

Firmware Architecture and Technical Context

Synth System Flow



Working engineering summary based on the supplied original and revised schematics, meeting notes, and system-flow discussion.

Document status

This report separates confirmed schematic observations from implementation assumptions. Exact clock values, device operating modes, protocol frames, and firmware policies must be verified against the final BOM, netlist, hardware configuration, and team requirements.

1. Executive Summary

The system modernises a classic multi-channel sampled-synthesis architecture. The original design used discrete counters, latches, waveform RAM, DACs, envelope logic, and analogue processing. The revised design transfers most waveform playback and timing behaviour into firmware running on eight dedicated STM32H725 channel processors, while retaining an analogue per-channel signal path and using a separate STM32H7-based effect processor for capture, mixing, effects, external input processing, and stereo output.

- Eight identical physical voice channels, each with its own STM32H725.
- Sample data is held in channel-accessible memory, most plausibly external SDRAM shown on the channel processor schematic.
- Each channel renders a continuous fixed-rate PCM stream, apparently targeting 192 kHz, and supplies it to its DAC through DMA.
- The analogue channel path includes DAC conversion and VCF/VCA/filter processing before the channel signal reaches the effect processor.
- Eight analogue channel outputs are digitised by two four-channel ADC stages and then mixed/processed in the effect processor.
- RS-485 is the likely real-time control bus. USB is the likely service/bulk-transfer path for sample loading, configuration, updates, and diagnostics.
- The stereo input path supports microphone/line input and processor-controlled 48 V phantom power.

Core firmware responsibility

The schematics define the electrical interfaces. The firmware must define sample management, voice state, playback rate, interpolation, DMA buffering, command protocol, scheduling, fault handling, effects, and safe power-control behaviour.

2. Original and Revised Architectures

2.1 Original channel-card system

The older drawings show a distributed hardware implementation of sample playback. Bus-control logic receives parameters, pitch/ramp circuitry generates waveform addresses, dedicated RAM holds waveform data, and multiplying DAC/analogue circuitry produces the envelope-controlled audio signal.

Original function	Modern equivalent
Pitch counters and address ramps	Fractional phase accumulator in firmware
Waveform RAM addressing logic	Memory reads from SDRAM/SRAM
Discrete envelope/control logic	Software envelope and modulation generators
Variable-rate DAC feeding	Fixed-rate PCM rendering plus interpolation
Hardware state machines	RTOS tasks, interrupts, and state machines

2.2 Revised system

The revised system uses eight digital channel processors and one effect processor. Each voice channel independently renders one voice, converts it to analogue, and sends its analogue output onward. The effect processor converts the eight channel signals back to digital, performs shared processing and mixing, and generates the final stereo output.

3. Complete System Architecture

3.1 Control and allocation

1. A key or UI event reaches the main controller.
2. The RTOS performs voice allocation and chooses one of the eight physical channels.
3. A compact command is sent over RS-485 to the selected channel MCU.
4. Sample/configuration data is loaded or serviced through the USB path as required.

3.2 One channel path - repeated eight times

5. Sample data available in channel SDRAM or other directly accessible memory.
6. STM32H725 maintains playback phase, pitch increment, loop state, envelope, and channel controls.
7. Renderer generates the next block of PCM at the fixed channel output rate.
8. PCM is written into the free half of a DMA ping-pong buffer.
9. DMA feeds the serial audio peripheral and DAC continuously.
10. DAC output passes through the analogue VCF/VCA/filter path.
11. Result appears as AUDIO_CH1 through AUDIO_CH8.

3.3 Effect and output path

12. The eight analogue channel signals are split across two four-channel ADC paths.
13. The effect MCU receives the digitised channel streams, de-interleaves them, and applies gain, pan, mixing, and effects.
14. The stereo result is sent to a stereo DAC and then to left/right line outputs.
15. Mic/line input is preamplified, digitised, and injected into the effect-processing path.

4. Communications and Data Movement

4.1 RS-485

RS-485 should be treated as the deterministic real-time command and status network, not as the continuous audio transport. Suitable messages include:

- NOTE_ON / NOTE_OFF
- voice assignment and sample ID
- pitch, velocity, loop, and envelope parameters
- filter/VCA control values
- mute, reset, diagnostics, and health/status

4.2 USB

USB is the natural service and bulk-data path. Depending on the final system design, it can support sample upload, configuration, firmware update, diagnostics, and possibly host audio or storage access. The precise USB topology and ownership still require confirmation.

4.3 Serial audio and analogue boundaries

Inside each channel, generated PCM is moved by DMA into the DAC interface. After DAC conversion, the channel passes through analogue processing. The effect processor therefore receives eight analogue channels, not eight direct PCM streams from the channel MCUs.

5. Sample Rates, Octaves, and Interpolation

The meeting notes show the octave-related legacy rates 7.5 kHz, 15 kHz, and 30 kHz. Each doubling corresponds to one octave. The notes also include the comparison $192 / 30 \approx 6.4$, which strongly suggests a fixed 192 kHz modern rendering domain used to represent the original variable-rate playback behaviour.

Legacy rate	192 kHz / rate	Meaning
7.5 kHz	25.6	About 25.6 output intervals per source interval
15 kHz	12.8	About 12.8 output intervals per source interval
30 kHz	6.4	About 6.4 output intervals per source interval

Important verification point

192 kHz is a strong interpretation of the notes and architecture, not yet a final firmware constant. Confirm the required SAI/DAC/ADC clocking and the exact legacy master-clock-derived rates before tuning or committing protocol formats.

5.1 Fixed-rate renderer

The renderer should use a fractional source position. At each 192 kHz output frame it reads neighbouring source samples, interpolates, applies envelope/modulation, writes the resulting PCM value, and advances the phase by the required increment.

phaseIncrement = sourcePlaybackRate / outputRate

A fixed-point accumulator is appropriate for deterministic real-time operation. Linear interpolation is the safest first implementation; higher-order interpolation can be evaluated later against both sound quality and vintage authenticity.

6. Sample Memory and Output DMA

6.1 Three distinct storage roles

Area	Purpose	Lifetime
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Master sample storage	Persistent sample library and assets	Long-term
Channel playback memory	Sample currently used by one channel, likely SDRAM	Per load / instrument
Output DMA buffer	Next few hundred rendered PCM frames	Milliseconds

6.2 DMA ping-pong model

Use a circular double buffer. DMA transmits one half while the audio task renders the other. Half-transfer and transfer-complete events should notify a high-priority render task rather than performing all DSP inside the interrupt.

Half-buffer	At 48 kHz	At 192 kHz
128 frames	2.667 ms	0.667 ms
256 frames	5.333 ms	1.333 ms

A 256-frame half-buffer is a reasonable bring-up starting point at 192 kHz. The final size should be selected after measuring worst-case render time, bus contention, interrupt latency, and end-to-end response.

6.3 STM32H7 cache and memory constraints

- Place buffers in DMA-accessible memory.
- Align DMA buffers to cache-line boundaries.
- Use a non-cacheable region or explicit D-cache clean/invalidate operations.
- Confirm the selected DMA engine can access the chosen SRAM/SDRAM region.
- Track underruns, missed deadlines, and maximum rendering time.

7. Firmware Responsibilities

7.1 Channel processor firmware

- board clocks, GPIO, SDRAM, DAC/SAI, DMA, watchdog and diagnostics
- RS-485 command parser and channel-addressing logic
- sample loader, sample descriptors, loop metadata and validation
- phase accumulator, interpolation, pitch mapping and tuning
- ADSR/envelope and modulation generation
- DMA-buffer rendering and underrun recovery
- analogue filter/VCA control outputs
- status reporting, fault counters and safe mute/reset behaviour

7.2 Effect processor firmware

- initialise and synchronise two four-channel ADC paths
- capture/de-interleave eight channel streams using DMA
- per-channel gain, pan, mute and effects send
- mix bus and shared effects processing
- mic/line ADC processing and routing
- stereo DAC output buffering

- 48 V phantom-control state machine
- clock, overflow, clipping, communication and power-fault monitoring

8. Phantom Power Control

The schematics include processor-facing 48 V control/status signals. Firmware should default phantom power off and use a sequenced state machine to avoid pops, protect connected equipment, and detect power faults.

16. Mute input path.
17. Assert 48 V enable.
18. Wait for 48 V power-good with timeout.
19. Apply settling delay.
20. Unmute input.
21. On disable: mute, remove 48 V, wait for discharge, then restore the permitted input state.

9. Recommended Bring-up Sequence

22. Clock tree and debug console
23. GPIO, reset lines and power-good inputs
24. External SDRAM test
25. DAC/SAI fixed test tone without RTOS complexity
26. Circular DMA and underrun instrumentation
27. Stored waveform at unity playback rate
28. Fractional phase and linear interpolation
29. 192 kHz performance profiling
30. RS-485 framing and channel addressing
31. Note-on/note-off and voice state
32. ADSR and filter/VCA control
33. Sample loading over USB/service path
34. Effect ADC capture and stereo pass-through
35. Eight-channel mix and effects
36. Mic/line input and safe phantom control
37. Full-system latency, tuning and soak tests

10. Suggested Software Model

10.1 Channel voice state

- SampleDescriptor: base address, length, source rate, loop start/end, format
- VoiceState: active, phase, phase increment, gain, envelope state, flags
- ChannelControl: pitch, velocity, filter, VCA, mute, diagnostics
- AudioBuffer: aligned two-half PCM output buffer

10.2 Minimal command set

Command	Typical content	Purpose
PING / STATUS	channel ID, version, faults	discovery and health

LOAD / SELECT	sample ID, address/length	prepare waveform
NOTE_ON	note, velocity, phase increment	start voice
NOTE_OFF	release mode	enter release stage
SET_PARAM	parameter ID and value	realtime control
MUTE / RESET	flags	safe recovery

11. Items Still Requiring Confirmation

- Exact channel DAC model, serial format, word width, master/slave clocking, and required fixed sample rate.
- Exact ADC operating rate and TDM/I2S topology in the effect processor.
- Which controller owns the master sample library and how data reaches channel SDRAM.
- Whether complete samples are loaded, streamed in blocks, or cached.
- Precise RS-485 physical topology, baud rate, addressing, framing, timing, and retry policy.
- USB role: host/device, connector ownership, class, and whether USB carries storage, service, or audio.
- Exact legacy playback-clock values and tuning constants rather than rounded 7.5/15/30 kHz values.
- Interpolation requirement: zero-order hold, linear, cubic, polyphase, or selectable vintage/clean modes.
- Analogue control interface and calibration required for VCF/VCA behaviour.
- Latency budget, startup sequencing, muting requirements, and acceptable fault recovery behaviour.

12. Working End-to-End Flow

KEY / UI -> MAIN CTRL -> RS-485 VOICE COMMAND

SAMPLE MEMORY -> STM32H725 RENDER -> DMA -> DAC -> VCF/VCA

8 ANALOG CHANNELS -> 2 x 4-CH ADC -> FX MCU -> STEREO DAC -> L/R OUT

Recommended first deliverable

Build one channel as a measurable real-time audio endpoint: SDRAM test, 192 kHz test tone, DMA ping-pong output, waveform playback, phase interpolation, RS-485 note command, and underrun telemetry. Once stable, replicate across all eight channels and integrate the effect processor.